OpsPilot — Video Demo & Narration Script

CentrAlign AI Engineering Hiring Assignment · Autonomous Operations Worker

Presenter: Syam Tammireddi	**Target Duration:** 2.5 – 3.0 Minutes	**Active Model:** Gemini 3.5 Flash (Live)	**Dashboard:** http://localhost:5173

🛠️ Pre-Recording Checklist (Before Clicking Record)

- Open browser to **http://localhost:5173** (or public tunnel link).
- Click **"Reset Sandbox"** (top-right circular button) to guarantee a clean baseline state.
- Verify top bar indicates **Model: Gemini 3.5 Flash (Live)** with a glowing green dot.
- Launch Loom or OBS screen recorder with microphone enabled.

Scene 1: Introduction & Architecture `0:00 – 0:30`

👉 **Action / Mouse Pointer:** Keep mouse in the top bar. Hover over "OpsPilot: Autonomous AI Worker" and "Model: Gemini 3.5 Flash (Live)".

👀 **Observed on Screen:** Dark-mode 3-panel UI. Left: Objective & Scenarios. Center: Execution Trace (IDLE). Right: ERP Ledger, Document Vault, Slack Hub.

🗣️ SPOKEN SCRIPT

"Hi everyone, my name is Syam Tammireddi, and today I'm demonstrating OpsPilot, an autonomous enterprise AI employee built for the CentrAlign AI engineering evaluation.

Traditional enterprise automation relies on rigid RPA scripts that break when APIs time out, or simple chatbots that can't execute real database writes.

*OpsPilot is an autonomous operations worker built completely from scratch in TypeScript—without LangChain or CrewAI—running live on Google Gemini 3.5 Flash. It follows the complete **Goal** → **Understand** → **Plan** → **Execute** → **Observe** → **Adapt** → **Verify** lifecycle across documents, ERP ledgers, and communications channels."*

Scene 2: Standard Autonomous Flow ($4,200 Invoice) `0:30 – 1:15`

👉 **Action / Buttons to Press:** 1. Click button: **[1. Standard Autonomous Invoice]** on left.
2. Click the blue button: **[Run Autonomous Worker]**.

👀 **Observed on Screen:** Steps 1 to 8 stream live in center trace. Every step has a green `ALLOW` shield. Step 4 calls `read_company_policy`. In right panel, ERP Ledger updates: INV-2024-001 turns to `PAID ($4,200.00)` with a bank reference. Top badge turns green: `COMPLETED & VERIFIED`.

🗣️ SPOKEN SCRIPT

"Let's trigger our first scenario with a natural language goal: 'Process the latest unpaid vendor invoice, follow company payment policy, and complete payment if authorized.'

Watch the live execution trace stream step-by-step: In Steps 1 & 2, the agent calls `search_documents` and `read_document`, discovering an unpaid invoice from Acme Cloud for $4,200. In Step 3, it queries our NetSuite ERP ledger to confirm it hasn't already been paid. In Step 4, the worker independently consults our internal company policy manual to discover spending rules. It reads that invoices under $5,000 can be paid autonomously. In Steps 5 and 6, it registers the invoice in the ERP and dispatches an ACH bank transfer. In Steps 7 and 8, it verifies ledger status and alerts the #finance-approvals Slack channel.

*Notice the green **ALLOW** badge on each step—every action was authorized by an independent Policy Engine before execution."*

Scene 3: Chaos Injection & Autonomous Failure Recovery

1:15 – 1:55

👉 **Action / Buttons to Press:** 1. Click top-right **[Reset Sandbox]**.
2. Click button: **[2. Failure Injection & Recovery]** (Failure Mode switches to `INJECTED (503)`).
3. Click blue button: **[Run Autonomous Worker]**.

👀 **Observed on Screen:** Step 6 `execute_payment` turns RED with `GATEWAY_TIMEOUT (503)`. Step 7 immediately adapts with decision: *'Payment failed with retryable timeout. Verifying zero debits and re-executing transaction.'* Step 7 re-calls `execute_payment` and turns GREEN (SUCCESS)! Finishes with `COMPLETED & VERIFIED`.

🗣️ SPOKEN SCRIPT

"Now let's test failure recovery, which is critical for CentrAlign's Observe → Adapt requirement. I've enabled failure injection mode, which simulates a bank payment gateway network partition.

Look at Step 6: When the agent attempts payment settlement, the gateway returns a 503 Gateway Timeout with `retryable: true`.

Instead of crashing or failing silently, OpsPilot observes the structured error from the sandbox, confirms that no funds were debited, and autonomously adapts by issuing an immediate retry.

The second payment succeeds, the ledger is updated, and the task completes successfully."

Scene 4: High-Value Human Approval Gate ($12,500 Invoice) 1:55 – 2:35

👉 **Action / Buttons to Press:** 1. Click top-right **[Reset Sandbox]**.
2. Click button: **[3. High-Value Approval Gate]**.
3. Click blue button: **[Run Autonomous Worker]**.
4. Wait 2 seconds → Agent pauses in `WAITING_FOR_APPROVAL`.
5. On right panel, click green button: **[Approve & Authorize Settlement]**.

👀 **Observed on Screen:** Trace halts at Step 6 with amber badge `WAITING_FOR_APPROVAL`. High-visibility red Approval Card appears: Snowflake ($12,500.00). When Approve is clicked, loop instantly unfreezes, settles payment, and completes with `COMPLETED & VERIFIED`.

🗣️ SPOKEN SCRIPT

"Next: what happens when an action exceeds autonomous authority?

Here the agent discovers a $12,500 invoice for Snowflake. When it reads company policy, it discovers that any transaction equal to or exceeding $5,000 strictly requires Finance Director sign-off.

The independent Policy Engine intercepts the proposed payment, marks it `REQUIRE_APPROVAL`, and halts the loop. The LLM cannot bypass this.

*The UI surfaces this manager approval card citing the exact policy violation. As Finance Director, I click **[Approve]**.*

The exact same agent loop resumes, executes the disbursement, and commits the transaction to the ledger."

Scene 5: Independent Verification & Closing

2:35 – 3:00

👉 **Action / Mouse Pointer:** Point mouse to the **Independent Verification Dossier** in the upper right column. Hover over the 5 green checkmarks.

👀 **Observed on Screen:** Box titled **Independent Verification Dossier (VERIFIED)** with 5/5 passed checks: ERP Ledger Existence, Settlement Status Verification (PAID), Bank Settlement Reference, Amount Integrity, and Stakeholder Notification.

🗣️ SPOKEN SCRIPT

*"Finally, look at this **Independent Verification Dossier**.*

We never rely on the LLM's own claim or a tool's 200 OK status code as proof of success. Our standalone Verification Engine queries the actual database to confirm five ground-truth checks: ledger existence, settlement status, valid bank reference, exact line-item amount matching, and audit delivery.

All five checks pass, producing this tamper-evident audit record.

That is OpsPilot: a dependable, policy-governed, self-healing AI employee for enterprise operations. Thank you!"

Submission Links:

- **Live Demo:** https://litigation-bulletin-cartridges-symbols.trycloudflare.com
- **GitHub Repo:** https://github.com/syamtammireddy/ops-pilot
- **Local UI:** http://localhost:5173